SOLUTIONS: TANGENTS, COTANGENTS, DERIVATIONS, AND DIFFERENTIALS

(1) **Motivating example**. Let $k = \mathbb{C}$, and let $A = k[x,y]/\langle x^2 - y^3 \rangle$. Then $\mathrm{Spec}(A)$ is the *cuspidal cubic*, and we can visualize it as (the closed points of) $\mathbb{V}(\langle x^2 - y^3 \rangle) \subseteq \mathbb{A}^2_k$.

(a) For any maximal ideal $\mathfrak{m} \neq \langle x,y \rangle$, show that $\mathfrak{m}/\mathfrak{m}^2 \cong k$ but $\langle x,y \rangle / \langle x,y \rangle^2 \cong k^2$

(b) Let $f = x^2 - y^3$, with formal partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$. For a point p lying on the curve in k^2, what is the rank of the Jacobian matrix $J(p) = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix}$?

(a) $\langle x,y \rangle / \langle x,y \rangle^2$ is spanned by x and y. The only relation we could impose is $x^2 = y^3$, but both of these live in $\langle x,y \rangle^2$ so don't do anything. Thus, $\langle x,y \rangle / \langle x,y \rangle^2$ is a 2-dimensional $\mathbb{C}$-vector space. Now, let $\mathfrak{m} = \langle x-a, y-b \rangle$ with a,b not both zero. To survive in the quotient we must additionally have $a^2 = b^3$, so in fact none of them can be zero (since if $a = 0$, so is b). Now, write $u = \overline{x} - a, v = \overline{y} - b$. Then $\mathfrak{m}/\mathfrak{m}^2$ is generated by u,v subject to $u^2 = v^2 = uv = 0$, $(u+a)^2 = (v+b)^3$. Expanding this out we get

$$u^2 + 2ua + a^2 = v^3 + 3v^2 b + 3vb^2 + b^3 \implies 2ua = 3vb^2$$

(since $a^2 = b^3$ and u^2, v^3 and $3v^2 b$ are all 0). Thus we can express $v = \frac{2a}{3b^2} u$, and so $\mathfrak{m}/\mathfrak{m}^2$ is in fact one-dimensional.

(b) The partial derivatives are respectively $2x$ and $-3y^2$, so the Jacobian matrix has rank zero if and only f $x = y = 0$

(2) **The Zariski Cotangent space**. Let R be a ring. For any closed point p of $\mathrm{Spec}(R)$, let $\kappa(p)$ denote the residue field.

(a) Let I be any ideal. Recall (or show) that for any R-module M, $M/IM \cong M \otimes_R R/I$, and thus $I/I^2 \cong I \otimes_R R/I$. Show that this gives I/I^2 the structure of an R/I module.

(b) Suppose $p = [\mathfrak{m}]$ is a closed point of $\mathrm{Spec}(R)$. Show that $\mathfrak{m}/\mathfrak{m}^2$ is a vector space over $\kappa(p)$.

(c) Let $R_\mathfrak{m}$ be the localization at $\mathfrak{m}$. Show that $\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{m}R_\mathfrak{m}/(\mathfrak{m}R_\mathfrak{m})^2$

(d) Show that if $\mathfrak{m}$ is finitely generated, then $\mathfrak{m}/\mathfrak{m}^2$ is finite dimensional (Hint: Nakayama's lemma).

(a) This is in the 'Modules I' worksheet. The isomorphism is given by the map $M \otimes_R R/I \to M/IM$ sending $m \otimes \overline{r} \to [rm]$; if r and s differed by $i \in I$ then their difference $(r-s)m$ would be in IM and thus zero in the quotient. Now, set $I = M$, thus I/I^2 is an R/I-module

(b) Let $I = \mathfrak{m}$. Thus $\mathfrak{m}/\mathfrak{m}^2$ is an $R/\mathfrak{m}$ module, but $R/\mathfrak{m}$ is precisely the field $\kappa(p)$, so $\mathfrak{m}/\mathfrak{m}^2$ is a $\kappa(p)$-vector space

(c) Let κ denote $\kappa(p) = R/\mathfrak{m}$. Now, $\mathfrak{m}/\mathfrak{m}^2$ has the structure of an $R/\mathfrak{m}$-module and is $R/\mathfrak{m} \otimes_R \mathfrak{m}$, while $\mathfrak{m}R_\mathfrak{m}/(\mathfrak{m}R_\mathfrak{m})^2$ has the structure of an $R_m/\mathfrak{m}R_m \cong R/\mathfrak{m}$-module given as $R_m/\mathfrak{m}R_m \otimes_{R_\mathfrak{m}} \mathfrak{m}R_\mathfrak{m}$, which is just the localization of the first. Since $\mathfrak{m}_\mathfrak{m} \cong \mathfrak{m} \otimes_R R_\mathfrak{m}$ as an $R_\mathfrak{m}$ module, nothing inside the ideal changes so they are the same.

(d) Let $M = \mathfrak{m}R_\mathfrak{m}$. Nakayama's lemma over a local tells us that a set X is a generating set for M as an $R_\mathfrak{m}$-module if and only if its image $\overline{X}$ in $M/\mathfrak{m}M$ is a spanning set for the vector space $M/\mathfrak{m}M$. Thus, if $\mathfrak{m}/\mathfrak{m}^2 \cong R_\mathfrak{m}/(\mathfrak{m}R_\mathfrak{m})^2 = M/\mathfrak{m}M$ is finite-dimensional, using the 'only if' direction we get that $\mathfrak{m}$ is finitely generated.

(3) **Point derivations**. Let A be a k-algebra and $\mathfrak{m}$ a maximal ideal of A. Let D be a derivation at $\mathfrak{m}$, i.e. a k-linear map $A \to k$ such that $D(fg) = f(\mathfrak{m})D(g) + g(\mathfrak{m})D(f)$.

(a) Show that if $f \in k$ i.e. f is a "constant function," then $D(f) = 0$

(b) Show that if f is idempotent, i.e. $f^2 = f$, then $D(f) = 0$

(c) For a ring R, an R-algebra S, and an S-module M, we can define $\mathrm{Der}_R(S, M)$ to be the set of R-linear maps $D : S \to M$ such that $D(fg) = fD(g) + gD(f)$. Setting $R = k, S = A, M = k$, we can recover our point derivations. If $\kappa(\mathfrak{m}) = \kappa(\mathfrak{n}) = k$, is $\mathrm{Der}_k(A, k)$ at $\mathfrak{m}$ isomorphic to $\mathrm{Der}_k(A, k)$ at $\mathfrak{n}$?

(a) Since we can pull constants out, $D(f) = fD(1)$, so it sufficed to show $D(1) = 0$. Since $1 = 1 \cdot 1$, $D(1) = 1 \cdot D(1) + 1 \cdot D(1) = 2D(1) \implies D(1) = 0$

(b) If $b^2 = b$ then $D(b) = D(b^2) = bD(b) + bD(b) \implies (1 - 2b)D(b) = 0$. Since $(1 - 2b)^2 = 1 - 4b + 4b^2 = 1 - 4b + 4b = 1$, $(1 - 2b)$ is a unit, so $(1 - 2b)D(b) = 0$ can only mean $D(b) = 0$A

(c) At a k-rational maximal ideal $\mathfrak{m}$, k can be given an A-module structure by $f \cdot x = f(\mathfrak{m})x = (f \bmod \mathfrak{m})x$. Then, derivations defined in the 'general' sense are exactly our point derivations since $f \cdot D(g) + g \cdot D(f) = f(\mathfrak{m})D(g) + g(\mathfrak{m})D(f)$. The notation $\mathrm{Der}_k(A, k)$ is misleading: if $\mathfrak{m}$ and $\mathfrak{n}$ are two k-rational maximal ideals, this gives k *different* structures as a A-module. Since we will prove that the dimension of the space of derivations is the same as that of $\mathfrak{m}/\mathfrak{m}^2$, the example of the cuspidal cubic shows that $\mathfrak{m}$ and $\mathfrak{n}$ can give different dimensions to '$\mathrm{Der}_k(A, k)$.'

(4) **Point derivations are dual to Zariski cotangents.** Let A be a k-algebra and $[\mathfrak{m}]$ a k-rational closed point i.e. $A/\mathfrak{m} = k$.

(a) Show that $A \cong k \oplus \mathfrak{m}$ as k-vector spaces.

(b) Recall (or show) the factoring-through principle: If $\phi : V \to W$ is a linear map of k-vector spaces and $\pi : V \to U$ is a surjective linear map such that $\ker \pi \subseteq \ker \phi$, then ϕ factors uniquely through π, i.e. there exists a unique map $\psi : U \to W$ such that $\phi = \psi \circ \pi$:

$$\begin{array}{ccc} V & \xrightarrow{\phi} & W \\ \downarrow{\scriptstyle \pi} & \nearrow_{!\psi} & \\ U & & \end{array}$$

(c) Show that given a derivation $D : A \to k$ at $\mathfrak{m}$, there is a unique way of associating to it an element of $(\mathfrak{m}/\mathfrak{m}^2)^\vee$, and that this association is linear

(d) Complete the proof that $\mathrm{Der}_k(A,k) \cong (\mathfrak{m}/\mathfrak{m}^2)^\vee$ by showing the surjectivity of the above assignment

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(a) Consider the exact sequence

$$0 \to \mathfrak{m} \to A \to k \to 0,$$

which is an exact sequence of k-vector spaces (k-algebras, in fact) since it is A-linear. Because exact sequences of vector spaces always split, this means that $A \cong k \oplus \mathfrak{m}$ as vector spaces

(b) We proved this in Math 101 and the worksheet question states "Recall (or show)"; we recall.

(c) Consider the following diagram of k-vector spaces:

$$\begin{array}{ccc} A \cong k \oplus \mathfrak{m} & \xrightarrow{D} & k \\ \downarrow{\scriptstyle \pi} & & \\ \mathfrak{m}/\mathfrak{m}^2 & & \end{array}$$

Here, π sends (a, f) to $[f]$ where $[f]$ represents the equivalence class of f under the projection $\mathfrak{m} \to \mathfrak{m}/\mathfrak{m}^2$. This is surjective, so to show that D factors through π we need to show that $\ker \pi \in \ker D$. The kernel of π is the subspace spanned by the constants and by the elements of $\mathfrak{m}^2$. We already know that derivations kill constants. Now, let $f, g \in \mathfrak{m}$, so $f(\mathfrak{m}) = g(\mathfrak{m}) = 0$, then $D(fg) = f(\mathfrak{m})D(g) + g(\mathfrak{m})D(f) = 0$. Thus the map factors through. This association is injective: if the associated map ψ is zero, then $D = \psi \circ \pi$ is also zero.

(d) We have a linear map $\mathfrak{m}/\mathfrak{m}^2 \to k$, and we want to construct a derivation from it. Define $\widetilde{\psi}$ to be the induced map on $\mathfrak{m}$ that kills $\mathfrak{m}^2$. Since $A = k \oplus \mathfrak{m}$ as k-vector spaces, we can write any $f \in A$ as $f(\mathfrak{m}) + \overline{f}$, where $\overline{f} \in \mathfrak{m}$. Define $D_\psi(f) = \widetilde{\psi}(\overline{f})$. This is clearly linear. Now suppose $f, g \in A$, then $fg = (f(\mathfrak{m}) + \overline{f})(g(\mathfrak{m}) + \overline{g}) = f(\mathfrak{m})g(\mathfrak{m}) + \overline{f}g(\mathfrak{m}) + \overline{g}f(\mathfrak{m}) + \overline{f}\overline{g}$. This $\overline{fg} = \overline{f}g(\mathfrak{m}) + \overline{g}f(\mathfrak{m}) + \overline{f}\overline{g}$. Since

$\overline{f}\overline{g} \in \mathfrak{m}^2$, $\widetilde{\psi}$ kills it, $D_\psi(fg) = \widetilde{\psi}(\overline{f}g(\mathfrak{m}) + \overline{g}f(\mathfrak{m})) = g(\mathfrak{m})\widetilde{\psi}(\overline{f}) + f(\mathfrak{m})\widetilde{\psi}(\overline{g}) = f(\mathfrak{m})D_\psi(g) + g(\mathfrak{m})D_\psi(f)$. Thus D_ψ is a derivation, and $\psi \circ \pi = D_\psi$.

(5) **Tangents and the dual numbers**. Let k be a field and R a k-algebra. Let $\pi, \psi : k[\epsilon]/\langle\epsilon\rangle^2 \to k$ be the k-linear maps given by $\pi(a+b\epsilon) = a, \psi(a+b\epsilon) = b$. Let $\phi : R \to k[\epsilon]/\langle\epsilon\rangle^2$ be a ring homomorphism and $\phi^\# : \mathrm{Spec}(k[\epsilon]/\langle\epsilon\rangle^2) \to \mathrm{Spec}\, R$ be the induced map on spectra.

(a) Show that $\phi^\#(\langle\epsilon\rangle)$, the image of the unique point in $\mathrm{Spec}(k[\epsilon]/\langle\epsilon^2\rangle)$, is a maximal ideal $\mathfrak{m}$ with residue field k

(b) Show that $\psi \circ \phi$ is a k-linear derivation of R at $p = [\mathfrak{m}]$

(c) Show that given a k-rational point p and a derivation D of R at p, there is a ring homomorphism from R to $k[\epsilon]/\langle\epsilon\rangle^2$

(d) Conclude that maps from $\mathrm{Spec}(k[\epsilon]/\langle\epsilon^2\rangle) \to \mathrm{Spec}(R)$ (induced by a ring map) are precisely the data of a k-rational point of $\mathrm{Spec}(R)$ along with a tangent vector at that point.

(a) Define $\phi_1 = \pi \circ \phi$ and $\phi_2 = \psi \circ \phi$ i.e. projections onto the first and second coordinates. By definition, $\phi^\#(\langle\epsilon\rangle) = \phi^{-1}(\langle\epsilon\rangle)$, so these are the elements without a first coordinate, i.e. $\ker \phi_1$. Since ϕ_1 is a map $R \to k$, its kernel is maximal with residue field k

(b) Let $f, g \in R$, then

$$\phi(fg) = (\phi_1(f) + \phi_2(f)\epsilon)(\phi_1(g) + \phi_2(g)\epsilon) = \phi_1(f)\phi_1(g) + (\phi_1(f)\phi_2(g) + \phi_1(g)\phi_2(f))\epsilon$$

where we have used $\epsilon^2 = 0$. Thus $\phi_2(fg) = \phi_1(f)\phi_2(g) + \phi_1(g)\phi_2(f)$. Since $\phi_1(f) = f(\mathfrak{m})$ and $\phi_1(g) = g(\mathfrak{m})$ by the previous problem, this can be written as $f(\mathfrak{m})\phi_2(g) + g(\mathfrak{m})\phi_2(f)$. Thus ϕ_2 is a derivation at $\mathfrak{m}$

(c) Define $\phi(f) = f(\mathfrak{m}) + D(f)\epsilon$. This is a valid ring homomorphism becasue it is clearly linear, send 0 to 0, and

$$\phi(fg) = (fg)(\mathfrak{m}) + D(fg)\epsilon = f(\mathfrak{m})g(\mathfrak{m}) + (f(\mathfrak{m})D(g) + g(\mathfrak{m})D(f))\epsilon = \phi(f)\phi(g)$$

since $\epsilon^2 = 0$

(d) Putting the previous parts together tells us that there is a bijective correspondence between ring maps $R \to k[\epsilon]/\langle\epsilon^2\rangle$ (affine scheme maps $\mathrm{Spec}(k[\epsilon]/\langle\epsilon^2\rangle) \to \mathrm{Spec}\, R$) and choices of a k-rational maximal ideal of R together with a derivation at that ideal. This is precisely the data of a point and a tangent vector at that point.

(6) **Universal property of $\Omega_{S/R}$** Let R be a ring, S an R-algebra, an M an S-module.

(a) Show that $\mathrm{Der}_R(S,M)$ and $\Omega_{S/R}$ are S-modules and that $d : S \to \Omega_{S/R}$ is a derivation

(b) Prove the universal property of $\Omega_{S/R}$

(a) $\Omega_{S/R}$ is a module over S because it is defined as the module quotient of a free module over S, and d is a derivation because (by the imposed relations) it is defined to be a derivation. Now, for $\mathrm{Der}_R(S,M)$, even though a derivation is only R-linear and not S-linear, $\mathrm{Der}_R(S,M)$ is still an S-module because for two derivations D_1, D_2 the S-linear combination $aD_1 + bD_2$ is also a derivation

(b) Given a derivation D, define a homomorphism $\psi : \Omega_{S/R} \to M$ via $\psi(df) = D(f)$. This suffices because we have given a homomorphism on the generators of $\Omega_{S/R}$, and it satisfies all the relations in $\Omega_{S/R}$ because (for instance) $D(fg) = fD(g) + gD(f)$ and $d(fg) = f\,dg + g\,df$. This association is bijective because it has an inverse given by $D(f) = \psi(df)$, which is a derivation by the same argument as before run backwards

(7) **Differentials and localization.** We will show that Ω commutes with localization, in an intuitive way via the "quotient rule." Let S be an R-algebra and U a multiplicatively closed subset of S.

(a) For $u \in U$, use the product rule to determine what $d(1/u)$ should be without a fraction in the derivation

(b) Construct a map $\Omega_{U^{-1}S/R} \to U^{-1}S \otimes_S \Omega_{S/R} \cong U^{-1}\Omega_{S/R}$

(c) Construct the inverse map $U^{-1}\Omega_{S/R} \to \Omega_{U^{-1}S/R}$

(d) Conclude that $\Omega_{U^{-1}S/R} \cong U^{-1}S \otimes_S \Omega_{S/R}$

(a) Since $u \cdot \frac{1}{u} = 1$, $0 = \frac{1}{u}du + ud(1/u)$. Rearranging and using that u is a unit in $U^{-1}S$, we get $d(1/u) = -u^{-2}du$

(b) The map is exactly the process we did: send $d(s/u) \mapsto u^{-2}(uds - sdu)$. The map is linear by the quotient rule calculation above

(c) An element of $U^{-1}\Omega_{S/R}$ is a linear combination of $\frac{s}{u}d(f)$, which can be interpreted as $\frac{s}{u}d(f/1)$ in $\Omega_{U^{-1}S/R}$

(d) These maps are inverses, so $\Omega_{U^{-1}S/R} \cong U^{-1}S \otimes_S \Omega_{S/R}$

(8) **The Conormal sequence.** Consider the following situation: we have a ring map $R \to S$ and another map $S \to T$. This exhibits S and T as R-algebras and T as an S-algebra, giving us three modules of differentials: $\Omega_{S/R}$, $\Omega_{T/R}$ and $\Omega_{T/S}$. The latter two are T-modules, while the first is an S-module. To get a T-module out of it, we can consider $T \otimes_S \Omega_{S/R}$.

(a) Show that there is an exact sequence of T-modules

$$T \otimes_S \Omega_{S/R} \xrightarrow{c \otimes db \mapsto c db} \Omega_{T/R} \xrightarrow{dc \mapsto dc} \Omega_{T/S} \longrightarrow 0$$

This is known as the **relative cotangent sequence** or the **first fundamental exact sequence**. (Hint: $\Omega_{T/S}$ and $\Omega_{T/R}$ are generated by the same elements, but there are extra relations in $\Omega_{T/S}$ since the differential kills everything in S (not just everything in R). For any $b \in S$, $db = 0$, but these are exactly the generators of $\Omega_{S/R}$ as an S-module, hence $T \otimes_S \Omega_{S/R}$ as a T-module)

(b) Suppose the second map in the given situation was a surjective map $\pi : S \to T$ of R-algebras. Show that $\Omega_{T/S} = 0$

(c) Continuing from the previous part, show that there is a well-defined map $I/I^2 \to T \otimes_S \Omega_{S/R}$ given by $f \mapsto 1 \otimes df$

(d) Show that the cokernel of this map is precisely $\Omega_{T/R}$, with the map as given in the first fundamental exact sequence. Conclude that there is an exact sequence

$$I/I^2 \xrightarrow{f \mapsto 1 \otimes df} T \otimes_S \Omega_{S/R} \xrightarrow{c \otimes db \mapsto c db} \Omega_{T/R} \longrightarrow 0$$

This is known as the **conormal sequence**, or the **second fundamental exact sequence**

(a) We follow the hint. The map $\Omega_{T/R} \to \Omega_{T/S}$ is clearly surjective since they are generated by the dt for $t \in T$. The kernel of the map is exactly the S-linearity relations $db = 0$ for $b \in S$, so the kernel is generated by the db as an S-module, but these are precisely the elements of $\Omega_{S/R}$! Since $\Omega_{S/R}$ is generated by db as an S-module, $1 \otimes db$ generate $T \otimes_S \Omega_{S/R}$ as a T-module.

(b) If $\pi : S \to T$ is a surjective map of R-algebras then $\Omega_{T/S}$ is generated by elements of T but under S-linearity i.e. $ds = 0$ for s in the image of S. But every element of T is in the image of S, so $\Omega_{T/S} = 0$. This means that the relative cotangent sequence turns to

$$T \otimes_S \Omega_{S/R} \xrightarrow{c \otimes db \mapsto c db} \Omega_{T/R} \longrightarrow 0$$

(c) To show that the given map is well-defined, we need to check that it kills I^2. Let $f = gh$ for $g, h \in I$. Then $df = gdh + hdg$, so

$$1 \otimes df = 1 \otimes gdh + 1 \otimes hdg = g \cdot (1 \otimes dh) + h \cdot (1 \otimes dg) = \pi(g) \otimes dh + \pi(h) \otimes dg,$$

but $\pi(g) = \pi(h) = 0$, so $1 \otimes df = 0$.

(d) The kernel of the map $T \otimes_S \Omega_{S/R} \xrightarrow{c\otimes db \mapsto c\,db} \Omega_{T/R}$ is generated by those $b \in S$ such that $db = 0$ in T/R, with the relations of R-linearity and the Leibniz rule. This is only if $b = 0$, so in S only if $\pi(f) = 0$ i.e. $f \in I$. Thus the kernel is precisely the image of the map $f \mapsto 1 \otimes df$. Extending the relative cotangent exact sequence to the left we get

$$I/I^2 \xrightarrow{f\mapsto 1\otimes df} T \otimes_S \Omega_{S/R} \xrightarrow{c\otimes db\mapsto c\,db} \Omega_{T/R} \longrightarrow 0$$

(9) **Some useful computations.**

(a) Show that $\Omega_{k[x_1,\ldots,x_n]/k}$ is the free $k[x_1,\ldots,x_n]$-module with generators dx_i

(b) Let K be a separable algebraic extension of k. Show that $\Omega_{K/k} = 0$. This is the main reason we will often assume k is perfect or has characteristic zero.

(c) Show that $\Omega_{k(t_1,\ldots,t_d)/k}$ is a $k(t_1,\ldots,t_d)$-vector space of dimension d

(d) Let K be a separable transcendental extension of k. Assume the fact that there exists a **separating transcendence basis**, i.e. a set of algebraically independent $t_1,\ldots,t_d \in K$, such that $d = \mathrm{trdeg}_k(K)$ and K is algebraic over $K' = k(t_1,\ldots,t_n)$. Show that $\Omega_{K/k}$ is a K-vector space of dimension d.

(a) Since the x_i generate $S = k[x_1,\ldots,x_n]$ as a k-algebra, their differentials dx_i generate $\Omega_{S/k}$ as an S-algebra so there is a surjective map $\psi : S^n \to \Omega_{S/k}$ sending the ith basis vector to dx_i. Now, consider the derivation $\partial_i : S \to S$. By the universal property, there is a unique homomorphism $\overline{\partial}_i \Omega_{S/k} \to S$ for each i, and it sends $dx_i \to 1$ and dx_j for $j \neq i$ to 0. Putting all of these maps together, we get a map ϕ into S^n i.e.

$$\Omega_{S/k} \xrightarrow{\begin{pmatrix} \overline{\partial}_1 \\ \overline{\partial}_2 \\ \vdots \\ \overline{\partial}_n \end{pmatrix}} S^n$$

This is an inverse to ψ since $\psi \circ \phi$ sends each dx_i to dx_i and $\phi \circ \psi$ sends each basis vector to itself. Thus $\Omega_{S/k}$ is isomorphic to $S^n = k[x_1,\ldots,x_n]^n$ with the basis given by the dx_i

(b) Let $\alpha \in K$, then it satisfies the minimal polynomial $p(\alpha) = 0$ with coefficients in k such that $p'(\alpha) \neq 0$. Since $p(\alpha) \in K$, we can apply d to it, and by k-linearity and the usual power rule (which we can easily show applies to any derivation),we get $d(p(\alpha)) = p'(\alpha) d\alpha = 0$. Since $p'(\alpha) \neq 0$, $d\alpha$ must be 0. This works for any $\alpha \in K$, so $\Omega_{K/k} = 0$

(c) $A = k(t_1,\ldots,t_d)$ is simply the fraction field of $S = k[t_1,\ldots,t_d]$. Since $\Omega_{\cdot/k}$ commutes with localization, $\Omega_{A/k} \cong A \otimes_S \Omega_{S/k}$. We proved in the first part that $\Omega_{S/k} \cong S^d$, so $A \otimes_S \Omega_{S/k} \cong A \otimes_S S^d = A^d$. Thus $\Omega_{A/k} \cong A^d$.

(d) **Note**: Here we prove this for *finitely-generated* K over k; when K is the fraction field of a finitely-generated integral k-algebra this is indeed the case. This will let us assume that the extension K'/K is finite, not just algebraic. In the book, Eisenbud shows that Ω commutes with direct limits so we get this property for infinite extensions too, but we won't need that for this worksheet.

Let $k \subset K' = k(t_1,\ldots,t_d) \subset K$ be as in the problem, i.e. K' is a pure transcendental extension of k and K is a separable algebraic extension of K'. By problems b) and c), $\Omega_{K'/k} \cong K'^d$ and $\Omega_{K/K'} = 0$.

Putting this in the relative cotangent sequence gets us

$$K \otimes_{K'} K'^d \longrightarrow \Omega_{K/k} \longrightarrow 0$$

The object on the left is just K^d, and the map is sending $dt_i \to dt_i$. By exactness on the right this map is surjective. This tells us that $\Omega_{K/k}$ is generated by the images of the dt_i under the inclusion map. We will show that the dt_i are linearly independent, hence $\Omega_{K/k} \cong K^d$ with basis $dt_1, \ldots, dt_d$. By the proof of part a), this would follow if there was a map sending $dt_i \to 1$ and the rest to 0 for each i, i.e. a derivation $K \to K$ over k sending $t_i \to 1$ and every other t_j to 0.

Because we have a finite separable extension K/K', it is generated by a primitive element α, and any element x of K' can be written as $c + \sum_{i=1}^n b_i \alpha^i = p(\alpha)$. For any derivation D, $D(x) = p'(\alpha)D(\alpha) + D(c) + \sum_{i=1}^n D(b_i)\alpha^i$. If q is the minimal polynomial of α, by this logic $0 = q'(\alpha)D(\alpha) + D(c) + \sum_{i=1}^n D(b_i)\alpha^i$, so $D(\alpha) = \frac{D(c)+\sum_{i=1}^n D(b_i)\alpha^i}{q'(\alpha)}$. This means that given a derivation on K', there is exactly one way of extending it to K (by defining it on α as above), and so we can just extend the derivations ∂_i on K' that separate the t_i.

(10) **Cotangents.** Let A be an integral k-algebra and let $\mathfrak{m}$ be a k-rational maximal ideal.

(a) What do you get if you apply the conormal sequence to the surjection $\pi : A \to A/\mathfrak{m}$ of k-algebras?

(b) Show that the first map in this sequence is an injection. (Hint: this is equivalent to showing that $df \in \mathfrak{m}\Omega_{A/k} \iff f \in \mathfrak{m}^2$. Now recall the proof showing the Zariski cotangent space is dual to derivations.)

(c) Conclude that $\mathfrak{m}/\mathfrak{m}^2 \cong \Omega_{A/k}/\mathfrak{m}\Omega_{A/k}$

(a) The kernel of this is $\mathfrak{m}/\mathfrak{m}^2$ and $A/\mathfrak{m}$ is k, so the conormal sequence becomes

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow k \otimes_A \Omega_{A/k} \longrightarrow \Omega_{k/k} = 0$$

(b) The first map is a surjection by the exactness of the conormal sequence. To show it is an injection, we construct a left-inverse $k \otimes_A \Omega_{A/k} \cong \Omega_{A/k}/\mathfrak{m}\Omega_{A/k}$. Recall the derivation we constructed for the Zariski cotangent space proof: $D : A \to k$ sending $f = f(\mathfrak{m}) + \overline{f}$ to $\overline{f} \mod \mathfrak{m}^2$, which is a derivation to $\mathfrak{m}/\mathfrak{m}^2$. Thus it is represented by a linear map $\Omega_{A/k} \to \mathfrak{m}/\mathfrak{m}^2$ that kills $\mathfrak{m}\Omega_{A/k}$: given $g df$ for $g \in \mathfrak{m}$, $\overline{f} \in \mathfrak{m}$ so $g\overline{f} \in \mathfrak{m}^2$ thus $g\overline{f} \mod \mathfrak{m}^2 = 0$. It is a left-inverse because if $f \in \mathfrak{m}/\mathfrak{m}^2$ then df gets sent exactly to f.

(c) The map in the conormal sequence is an injection and a surjection, so $\mathfrak{m}/\mathfrak{m}^2 \cong \Omega_{A/k}/\mathfrak{m}\Omega_{A/k}$

Remark: if $\mathfrak{m}$ is not a k-rational maximal ideal, then the same computation goes through provided $\Omega_{\kappa(\mathfrak{m})/k} = 0$. As we saw, this always holds if the extension is separable but might not hold if the extension is not. Thus the equivalence between the Zariski cotangent space $\mathfrak{m}/\mathfrak{m}^2$ and the fiber $\Omega/\mathfrak{m}$ in general only holds for perfect fields. In the worksheet, I say smoothness and regularity are the same, but in fact smoothness is defined via Ω and regularity via $\mathfrak{m}/\mathfrak{m}^2$ (since that controls the number of elements needed to generate $\mathfrak{m}$, i.e. whether $R_{\mathfrak{m}}$ is regular or not). For perfect fields, by the results of this exercise, they are indeed the same.

(11) **The Jacobian matrix.** Let A be an integral k-algebra, and k be algebraically closed. It turns out that if we write $S = k[x_1, \ldots, x_n]$, and $A = S/I$ where $I = \langle f_1, \ldots, f_r \rangle$, the first map in the conormal sequence $I/I^2 \to A \otimes_{k[x_1,\ldots,x_n]} \Omega_{k[x_1,\ldots,x_n]/k} \to \Omega_{A/k} \to 0$ has a nice explicit description via the Jacobian matrix $J = [\partial f_i / \partial x_j]$, giving us an explicit description of $\Omega_{A/k}$. Tensoring with $k = A/\mathfrak{m}$ then lets us compute the dimension of $\mathfrak{m}/\mathfrak{m}^2$.

(a) Consider the conormal sequence associated to the surjection $S \to A$. What is $A \otimes \Omega_{S/k}$? Using this description, describe the map $I/I^2 \to A \otimes \Omega_{S/k}$

(b) Compute $\Omega_{A/k}$ for $A = k[x_1, \ldots, x_n]/\langle f_1, \ldots, f_r \rangle$

(c) Let $\mathfrak{m}$ be a maximal ideal and give k the structure of an A-module via the map $A \to A/\mathfrak{m}$. Tensor the conormal sequence we have with k to get a new sequence. Explain why it is exact, and what the terms and maps are.

(d) Prove the **Jacobian criterion for smoothness**: At a point $p = [\mathfrak{m}]$, the dimension of the Zariski cotangent space $\mathfrak{m}/\mathfrak{m}^2$ is the co-rank of the Jacobian matrix $J(p) = [\partial f_i / \partial x_j]|_p$.

(a) By problem 9, $\Omega_{S/k} = S^n$, so $\Omega_{S/k} \otimes_S A = A^n$. Thus the conormal sequence becomes

$$I/I^2 \longrightarrow A^n \longrightarrow \Omega_{A/k} \longrightarrow 0$$

The basis for A^k here is given by the dx_i, so we want to express the map $f \mapsto df$ in terms of this basis. We can write any $f \in I$ as a S-linear combination of f_i, say $\sum_i g_i f_i$. On applying d, thus turns to $\sum_i (g_i df_i + f_i dg_i)$, but the $f_i g d_i$ term is still in I so dies in A^k. Thus, $\sum_i g_i f_i \mapsto \sum_i g_i df_i$, and we know $df_i = \sum_j \frac{\partial f_i}{\partial x_j} dx_j$. Thus, the matrix for this transformation is exactly the Jacobian matrix $[\partial f_i / \partial x_j]$, noting that the elements in it are polynomials, not numbers (yet).

(b) By the conormal sequence, $\Omega_{A/k} \cong A_k / \mathrm{img}(J)$. Thus it is the quotient of the free A module on the dx_i by the relations posed by $df_i = \sum_j \frac{\partial f_i}{\partial x_j} dx_j = 0$

(c) Tensoring is a right-exact functor, so we can tensor the right-exact conormal sequence to get another right-exact sequence. Then, the middle term A^n turns to k^n and the map is the "evaluate at $\mathfrak{m}$" map. By our previous identification, $\Omega_{A/k} \otimes k$ is exactly $\mathfrak{m}/\mathfrak{m}^2$; thus $\mathfrak{m}/\mathfrak{m}^2$ is the cokernel of the matrix $J \otimes k$, which evaluates the polynomials at $\mathfrak{m}$ to give us a matrix of numbers. The effect of this operation is to evaluate polynomials, since for an R module, tensoring with $k = R/\mathfrak{m}$ is evaluation at $\mathfrak{m}$

(d) Since $\mathfrak{m}/\mathfrak{m}^2$ is the cokernel of the Jacobian, its dimension is the rank of the Jacobian. This is just a matrix of numbers whose rank is easily computed. Thus, if we have a description of our coordinate ring as the quotient of the polynomial algebra by $f_1, \ldots, f_r$, evaluating the Jacobian at a point tells us whether the associated variety is smooth at that point.

(12) **The dimension of a variety**. Let k be a perfect field and A a finitely-generated integral k-algebra of dimension d. Our goal here is to show that for any maximal ideal $\mathfrak{m}$ of A, $\mathrm{ht}(\mathfrak{m}) = d$, and that $\dim_k(\mathfrak{m}/\mathfrak{m}^2) \geq d$, justifying a 'local' check for smoothness.

(a) Let $B = k[y_1, \ldots, y_d]$ be a Noether normalization of A. Let $\mathfrak{m}$ is a maximal ideal of A. Justify the statement: $\mathfrak{m} \cap B$ is a maximal ideal of B with the same height as that of $\mathfrak{m}$

(b) Show that every maximal ideal in B has height d.

(c) Conclude that every maximal ideal of A has height equal to $\dim(A)$.

(d) Use Krull's height theorem to show that $\dim_{\kappa(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2 \geq \mathrm{height}(\mathfrak{m}) = \dim(A)$.

(e) Show that the variety associated to A is smooth at point $[\mathfrak{m}]$ if and only if $A_{\mathfrak{m}}$ is a regular local ring. Thus, even though smoothness was originally defined using a global quantity ($\dim A$), it can be checked locally

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(a) The map $B \to A$ is an integral extension, so $\mathfrak{m}$ is maximal if and only if $\mathfrak{m} \cap B$ is maximal (by Lying over, as in worksheet 7-2). Moreover, since B is normal, it satisfies the going down property and so (again, by a result we showed in worksheet 7-2) $\mathrm{height}(\mathfrak{m}) = (\mathrm{height}(\mathfrak{m} \cap B)$

(b) First suppose k was algebraically closed. Then $\mathfrak{m} = \langle x_1 - a_1, \ldots, x_d - a_d \rangle$, so by constructing the chain $\langle 0 \rangle \subsetneq \langle x_1 - a_1 \rangle \subsetneq \langle x_1 - a_1, x_2 - a_2 \rangle \subsetneq \cdots \subsetneq \mathfrak{m}$, we see that the height is at least d, but because the Krull dimension of B is d, it can be no more than d, so it is exactly d.

Now suppose k was not algebraically closed. Then, by applying the Nullstellensatz to $\overline{k}$, we know that there is a finite algebraic extension K of k with a maximal ideal $\overline{\mathfrak{m}} = \langle x_i - a_i \rangle$ (with $a_i \in K$) such that $\mathfrak{m} = \overline{\mathfrak{m}} \cap K[x_1, \ldots, x_d]$. This latter prime has height exactly d by the argument in the previous paragraph. Since K is an algebraic extension of k, $K[x_1, \ldots, x_d]$ is an integral extension of $k[x_1, \ldots, x_d]$ (for any polynomial, the coefficients are algebraic over k and thus $k[x_1, \ldots, x_d]$, and the variables are already in $k[x_1, \ldots, x_d]$) and $k[x_1, \ldots, x_d]$ is normal, by going down we get that the height of $\mathfrak{m}$ is exactly d.

(c) Putting the last two parts together, a maximal ideal $\mathfrak{m}$ in A lies over an ideal $\mathfrak{m} \cap B$ in the Noether normalization of A. By part a) this is maximal and has the same height as $\mathfrak{m}$, which by part b) is d. Thus every maximal ideal has the same dimension.

(d) Krull's height theorem says that if $\mathfrak{p}$ is a minimal prime over $\langle f_1, \ldots, f_r \rangle$, then $\mathrm{ht}(p) \leq r$. For a maximal ideal $\mathfrak{m}$, the minimum number of elements needed to generate $\mathfrak{m}$ (as an $R_{\mathfrak{m}}$-module) is exactly $\dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) = r$. Let $f_1, \ldots, f_r$ be such a generating set. Then since $\mathfrak{m} = \langle f_1, \ldots, f_r \rangle$, it is certainly a minimal prime over $\langle f_1, \ldots, f_r \rangle$, so $\mathrm{ht}(\mathfrak{m}) \leq r$. But we saw in the previous problem that the height of $\mathfrak{m}$ is the dimension of A. Thus $\dim(A) \leq \dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2)$.

(e) If A is smooth at $\mathfrak{m}$ then $\dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) = \dim(A) = \mathrm{ht}(\mathfrak{m})$. Consider the local ring $A_{\mathfrak{m}}$; it is regular if and only if $\dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) = \dim(A_{\mathfrak{m}}) = \mathrm{ht}(\mathfrak{m})$, since $\dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2)$ is the minimal number of generators of $\mathfrak{m}$. Technically, this is the only point in the problem where we have used perfectness of k, to use the equality of $\mathfrak{m}/\mathfrak{m}^2$ with $\Omega/\mathfrak{m}$.

(13) **Geometric Nakayama**. Let M be a finitely generated module over a ring R. Let $\mathfrak{p}$ be a prime of R such that the fiber of M at $\mathfrak{p}$, defined as $M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p}$, has dimension n as an $R_\mathfrak{p}/\mathfrak{p}R_\mathfrak{p}$ vector space

(a) Show that there exist n elements $f_1,\dots,f_n$ that generate $M_\mathfrak{p}$ as an $R_\mathfrak{p}$ module

(b) Show that there exists a g such that the images of the numerators of $f_1,\dots,f_n$ generate $M[1/g]$ as a $R[1/g]$-module.

(c) Show that for any $[\mathfrak{q}] \in D(g)$, these elements generate $M_\mathfrak{q}$ as a $R_\mathfrak{q}$ module

(d) Show that this means $\dim_{\kappa(\mathfrak{q})}(M_\mathfrak{q}/\mathfrak{q}M_\mathfrak{q})$ is less than or equal to n, for every $\mathfrak{q} \in D(g)$

(a) Since $R_\mathfrak{p}$ is a local ring, we know that the minimum number of generators for $M_\mathfrak{p}$ over $R_\mathfrak{p}$ is the dimension of $M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p}$ as a $\kappa(\mathfrak{p})$-vector space. Since this is n, there exist n elements $f_1,\dots,f_n \in R_\mathfrak{p}$ such that they generate $M_\mathfrak{p}$.

(b) Suppose $M_\mathfrak{p}$ is generated by $f_1,\dots,f_n \in R_\mathfrak{p}$. Then, we can write $f_i = m_i/u_i$ for $m_i \in M$, $u_i \notin \mathfrak{p}$. For any $m/u \in M_\mathfrak{q}$, there is a linear combination

$$\sum_{i=1}^{n} a_i \frac{m_i}{u_i} = \frac{m}{u} \implies \sum_{i=1}^{n} \frac{a_i}{u_i}\frac{m_i}{1}$$

i.e. the elements $m_i/1$ generate $M_\mathfrak{p}$ as an $R_\mathfrak{p}$ module. Define a map $\phi_\mathfrak{p} : R_\mathfrak{p}^n \to M_\mathfrak{p}$ that sends the ith basis vector to $m_i/1$. Then, this map is surjective, and moreover it is the localization of $\phi : R^n \to M$ that sends the ith basis vector to m_i. Since $\operatorname{coker}\phi$ is the quotient of a finitely-generated module M, it is finitely generated and satisfies $(\operatorname{coker}\phi)_\mathfrak{p} = \operatorname{coker}(\phi_\mathfrak{p}) = 0$. Now consider the boxed lemma:

> **Lemma**: Let N be a finitely-generated module over a ring R such that $N_\mathfrak{p} = 0$ for some prime ideal $\mathfrak{p}$. Then there exists some $g \in R$ such that $N[1/g] = 0$
>
> *Proof.* If $N_\mathfrak{p} = 0$ then the image of any $n \in N$ is 0 in $N_\mathfrak{p}$, so there is some $u \notin \mathfrak{p}$ such that $un = 0$. Now, let $n_1,\dots,n_t$ be a generating set for N, and $u_i \in \mathfrak{p}$ be such that $u_i n_i = 0$. Since $\mathfrak{p}$ is prime, $u = u_1 u_2 \cdots u_t \notin \mathfrak{p}$, and by construction u kills every n_i. Since the n_i generate, u kills every $n \in N$
>
> Now let $g = u$ and consider $N[1/g]$. For any $n \in N$, $gn = 0 \implies \frac{n}{1} = 0$ in $N[1/g]$ i.e. $N[1/g] = 0$ $\square$

By this lemma, there is some $g \in R$ such that $(\operatorname{coker}\phi)[1/g] = 0$, i.e. the map $\phi[1/g] : R[1/g]^n \to M[1/g]$ sending the ith basis vector to $\frac{m_i}{1}$ in $M[1/g]$ is surjective. Thus the numerators of $f_1,\dots,f_n$ generate $M[1/g]$ as a $R[1/g]$-module

(c) Consider the map ϕ from the previous part. We know that $(\operatorname{coker}\phi)[1/g] = 0$. Now, if $\mathfrak{q} \in D(g)$ then $g \notin \mathfrak{q} \implies (R \setminus \mathfrak{q}) \supseteq \{1, g, g^2, \dots\}$, i.e. $R_\mathfrak{q}$ is a further localization of $R[1/g]$. Thus, if $(\operatorname{coker}\phi)[1/g] = 0$ then $(\operatorname{coker}\phi)_\mathfrak{q} = 0 \implies \phi_\mathfrak{q} : R_\mathfrak{q}^n \to M_\mathfrak{q}$ is surjective $\implies M_\mathfrak{q}$ is generated by the images of $\widetilde{f_n},\dots,\widetilde{f_n}$

(d) If n elements generate $M_{\mathfrak{q}}$ as an $R_{\mathfrak{q}}$ module, then the minimum number of generators of $M_{\mathfrak{q}}$ is less than or equal to n. Since this quantity is $\dim_{\kappa(\mathfrak{q})}(M_{\mathfrak{q}}/\mathfrak{q}M_{\mathfrak{q}})$, this means that $\dim_{\kappa(\mathfrak{q})}(M_{\mathfrak{q}}/\mathfrak{q}M_{\mathfrak{q}}) \leq n$ for every $\mathfrak{q} \in D(g)$

(14) **Generic Smoothness.** Armed with the geometric Nakayama, we only need to do the calculation of the fiber dimension of Ω at one convenient point. Let A be a finitely-generated integral k-algebra and $\Omega = \Omega_{A/k}$ the module of differentials. Suppose that k has characteristic zero.

(a) Show that the fiber of Ω at $\langle 0 \rangle$ is $\Omega_{K/k}$, where $K = \mathrm{Frac}(A)$

(b) Show that $\Omega_{K/k}$ is a $\dim(A)$-dimensional vector space over K

(c) Use geometric Nakayama to conclude that there is a Zariski open set U such that for any maximal idea $\mathfrak{m} \in U$, $\dim(\mathfrak{m}/\mathfrak{m}^2)$ is exactly $\dim(A)$

(a) The local module $\Omega_{\langle 0 \rangle}$ is $\Omega \otimes_A A_{\langle 0 \rangle} = \Omega \otimes_A \mathrm{Frac}(A)$, which is a K-vector space and thus *is* the fiber (no further quotienting necessary). Since K is the localization of A at $\langle 0 \rangle$, and localization commutes with Ω, we get that $\Omega \otimes_A A_{\langle 0 \rangle} = (A \setminus 0)^{-1}\Omega_{A/k} \cong \Omega_{A_{\langle 0 \rangle}/k} = \Omega_{K/k}$

(b) Since k has characteristic 0, K is a separable transcendental extension of k. By problem 9d), $\Omega_{K/k}$ is a d-dimensional vector space over K, where d is the transcendence degree of K over k. But we know by Noether normalization that $\mathrm{trdeg}_k(K) = \dim(A)$. Thus $\Omega_{K/k}$ is a $\dim(A)$-dimensional vector space over A

(c) By geometric Nakayama, there exists a Zariski open set U such that for every prime $\mathfrak{p}$ in U, the fiber of Ω at $\mathfrak{p}$ is at most $\dim(A)$. For every maximal ideal $\mathfrak{m}$ in A, since $\Omega/\mathfrak{m} \cong \dim_{\kappa(\mathfrak{m})}(\mathfrak{m}/\mathfrak{m}^2) \geq \dim(A)$, we know that the fiber dimension is also at least $\dim(A)$. Thus at each such maximal ideal the dimension is exactly $\dim(A)$.